

Diverse Alpha Ensembles for A-Share Cross-Sectional Prediction

Correlation-Constrained Search, Explicit Residualization, and Hierarchical Nesting

YYB Quantitative Research Platform

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Abstract

We report an auditable exploratory search for A-share cross-sectional signals using a local panel with strict 2020–2022 in-sample construction and 2023 out-of-sample (OOS) measurement. The afternoon extension expands the original combination matrix with near-threshold residualization, cross-style bridge portfolios, explicit weighted style tilts, and hierarchical nesting. A hard screen identifies 10 displayed factors with OOS Sharpe greater than 8 and pairwise absolute daily-PnL correlation below 0.5; the maximum observed pairwise correlation is 0.4980. The displayed set contains 0 atomic factors and 10 measured ensembles. Because the extension adaptively inspects the 2023 OOS window, these results are an exploratory research artifact, not a deployable or publication-ready performance claim.

1 Research status

The artifact is designed to be honest about what is and is not established. It improves engineering reproducibility, economic lineage, and portfolio diversity. It does not yet meet a top-conference empirical standard: a new untouched holdout, walk-forward regime splits, transaction-cost modeling, capacity analysis, and formal multiple-testing control remain required.

2 Related work and positioning

The atomic primitives connect to established return continuation and reversal evidence [1, 2]. The cross-style ensembles combine momentum, liquidity, crowding, and nonlinear interactions in the spirit of modern empirical asset-pricing prediction [5]. Because the afternoon workflow adaptively compares many candidates, its reported Sharpe ratios require stronger safeguards than a conventional train-test split: factor-zoo multiple-testing discipline [4], White’s reality check [6], Hansen’s superior-predictive-ability test [7], and the deflated Sharpe ratio [8]. Trading-cost and capacity analysis is also essential before deployment [9]. The present artifact implements transparent lineage and a frozen next-holdout protocol; it does not claim those missing tests have already passed.

3 Protocol

Atomic factors are ranked using IS-only metrics. Ensemble construction standardizes each leaf cross-sectionally before aggregation. Residualization variants remove market-cap buckets, log-market-cap exposure, rolling beta, or joint log-market-cap plus beta exposure. Portfolio selection is a greedy hard screen:

$$\text{Sharpe}_i^{OOS} > 8, \quad \max_{j < i} |\text{Corr}(\Delta P_i, \Delta P_j)| < 0.5,$$

where P_i is cumulative 2023 OOS PnL and ΔP_i is its daily change.

4 Expanded innovation search

The afternoon extension executes sequentially in a single Flask pipeline to keep memory bounded. It records each real API result before proceeding. Search families include: residualized near-threshold candidates; cross-style bridges; style-specific anchors; nested style anchors; offset cross-style baskets; explicit weighted style tilts; pair/triple combo nesting; and L4 meta nesting.

Exploratory innovation phases: best measured OOS Sharpe

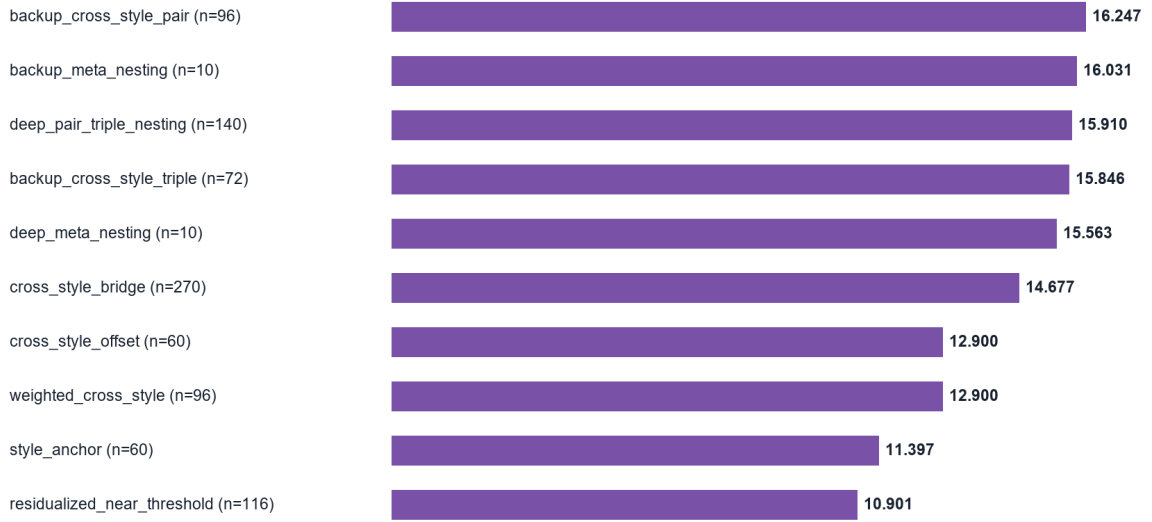


Figure 1: Best measured OOS Sharpe by exploratory innovation phase.

Table 1: Recorded exploratory experiments by phase.

Phase	Rows	Sharpe > 8	Median	P90	Max.
backup_cross_style_pair	96	96	12.398	14.796	16.247
backup_cross_style_triple	72	72	13.415	14.848	15.846
backup_meta_nesting	10	10	15.655	15.954	16.031
cross_style_bridge	270	253	10.016	11.545	14.677
cross_style_offset	60	48	9.617	12.632	12.900
deep_meta_nesting	10	10	15.396	15.550	15.563
deep_pair_triple_nesting	140	140	13.808	14.907	15.910
nested_style_anchor	20	15	9.287	10.691	10.799
residualized_near_threshold	116	78	8.200	10.227	10.901
style_anchor	60	33	8.102	10.745	11.397
weighted_cross_style	96	64	9.402	12.475	12.900

Table 2: Descriptive neutralization comparison across recorded experiments. Candidate mixtures differ by row.

Neutralization	Rows	Median	P90	Max.
beta	258	10.408	14.268	15.846
market_cap	168	11.514	14.514	16.247
market_cap_beta	258	10.373	14.101	15.945
market_cap_regression	37	8.597	12.592	16.031
none	229	10.656	14.310	15.816

Table 3: Greedy strict-pool sensitivity to the absolute daily-PnL correlation threshold.

Correlation threshold	Strict pool size
0.35	8
0.40	10
0.45	13
0.50	22
0.55	23

5 Descriptive sequential split audit

The following splits are descriptive diagnostics inside the same adaptively inspected 2023 OOS window, not a new untouched holdout. All ten displayed factors remain positive in both sequential half-year blocks. However, the full-window diversity screen is not uniformly stable across subperiods: the highest subperiod pairwise absolute daily-PnL correlation is 0.7409.

Table 4: Descriptive sequential splits within the inspected 2023 OOS window.

Segment	Days	Equal-weight Sharpe	Maximum pairwise corr.
Full 2023 OOS	241	21.269	0.4980
H1 sequential	120	17.312	0.5866
H2 sequential	121	27.135	0.6097
Q1 sequential	60	22.707	0.6489
Q2 sequential	60	16.422	0.6476
Q3 sequential	60	37.962	0.5311
Q4 sequential	61	22.891	0.7409

6 Strict final portfolio

Final strict portfolio: measured OOS Sharpe

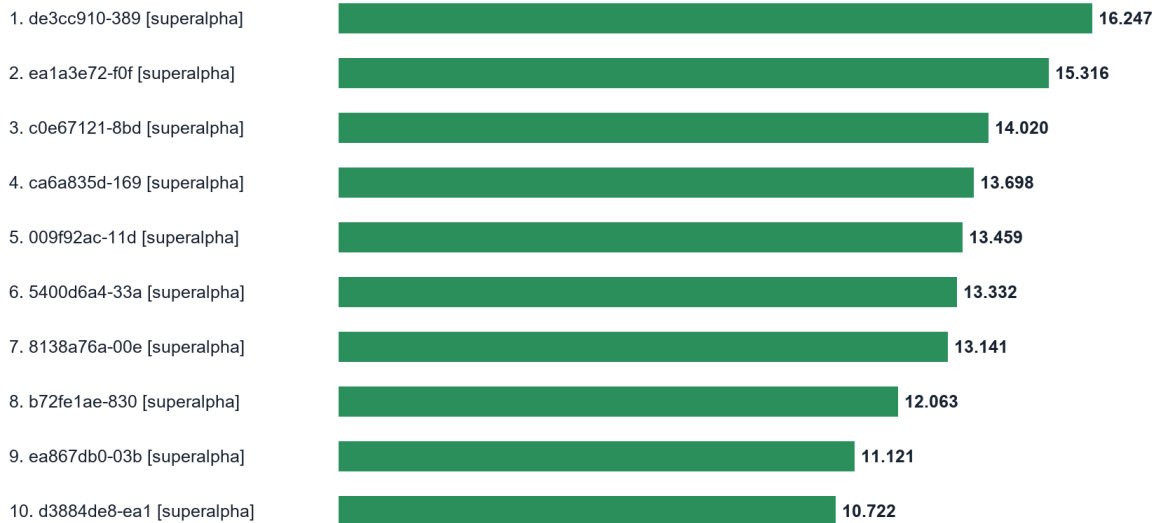


Figure 2: Measured OOS Sharpe for the displayed strict portfolio.

Table 5: Final displayed factors: hard Sharpe and daily-PnL correlation constraints.

Rank	ID prefix	Type	Theme	Sharpe	IC	Turnover	Max corr.
1	de3cc910-389	superalpha	composite	16.247	0.0463	0.4180	0.0000
2	ea1a3e72-f0f	superalpha	composite	15.316	0.0320	0.4791	0.4721
3	c0e67121-8bd	superalpha	composite	14.020	0.0376	0.6462	0.4830
4	ca6a835d-169	superalpha	composite	13.698	0.0349	0.4323	0.4876
5	009f92ac-11d	superalpha	composite	13.459	0.0447	0.5656	0.4980
6	5400d6a4-33a	superalpha	composite	13.332	0.0313	0.4649	0.4694
7	8138a76a-00e	superalpha	momentum	13.141	0.0418	0.5827	0.4959
8	b72fe1ae-830	superalpha	composite	12.063	0.0396	0.5954	0.4348
9	ea867db0-03b	superalpha	reversal	11.121	0.0300	0.7376	0.3834
10	d3884de8-ea1	superalpha	momentum	10.722	0.0372	0.4340	0.4828

Absolute daily-PnL correlation: final strict portfolio

	1	2	3	4	5	6	7	8	9	10
1	1.00	0.47	0.46	0.47	0.50	0.47	0.50	0.39	0.38	0.36
2	0.47	1.00	0.48	0.49	0.31	0.41	0.04	0.23	0.21	0.18
3	0.46	0.48	1.00	0.46	0.37	0.45	0.14	0.42	0.24	0.11
4	0.47	0.49	0.46	1.00	0.49	0.41	0.09	0.31	0.36	0.18
5	0.50	0.31	0.37	0.49	1.00	0.28	0.50	0.43	0.23	0.48
6	0.47	0.41	0.45	0.41	0.28	1.00	0.15	0.22	0.14	0.04
7	0.50	0.04	0.14	0.09	0.50	0.15	1.00	0.43	0.09	0.31
8	0.39	0.23	0.42	0.31	0.43	0.22	0.43	1.00	0.23	0.13
9	0.38	0.21	0.24	0.36	0.23	0.14	0.09	0.23	1.00	0.41
10	0.36	0.18	0.11	0.18	0.48	0.04	0.31	0.13	0.41	1.00

Figure 3: Absolute daily-PnL correlation matrix for the displayed factors.

Cumulative OOS PnL: final strict portfolio

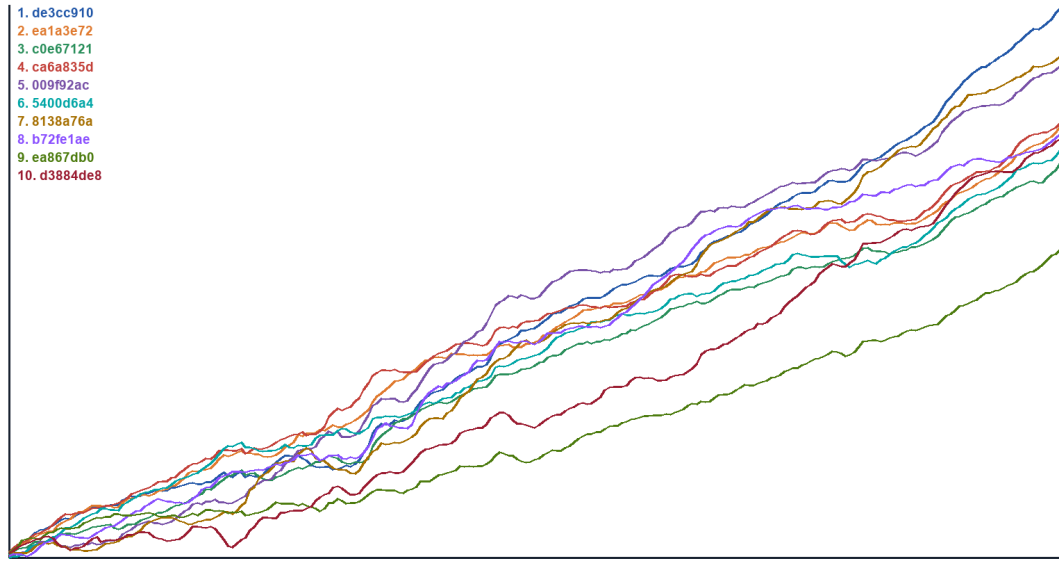


Figure 4: Cumulative 2023 OOS PnL curves for the displayed factors.

7 Economic interpretation

The table separates ensemble-level meaning from detected child proxies. New residualized and nested records preserve lineage through `superalpha_ref(...)` references. Historical cache placeholders remain explicitly marked rather than reverse-engineered.

Rank	ID prefix	Economic interpretation	Detected child proxies
1	de3cc910-389	liquidity and crowding; trend or continuation	amihud_20d, log_dollar_vol, ret_20d, turnover_rate
2	ea1a3e72-f0f	liquidity and crowding; short-horizon reversal	turnover_5d, abnormal_vol_rev, turnover_rate, amihud_20d, log_dollar_vol
3	c0e67121-8bd	liquidity and crowding; short-horizon reversal	turnover_rate, auction_return, amihud_20d, log_dollar_vol
4	ca6a835d-169	liquidity and crowding; trend or continuation; short-horizon reversal	turnover_rate, auction_return, ret_120d_skip5, log_dollar_vol, amihud_20d
5	009f92ac-11d	liquidity and crowding; trend or continuation	ret_20d, turnover_rate, ret_120d_skip5, log_dollar_vol
6	5400d6a4-33a	liquidity and crowding; trend or continuation; short-horizon reversal	ret_20d, volume_profile_ratio, ret_120d_skip5, abnormal_vol_rev, turnover_5d, amihud_20d, log_dollar_vol, rev_5d
7	8138a76a-00e	liquidity and crowding; trend or continuation	ret_20d, turnover_rate
8	b72fe1ae-830	liquidity and crowding; short-horizon reversal	turnover_5d, abnormal_vol_rev, turnover_rate, auction_return, volume_profile_ratio, amihud_20d
9	ea867db0-03b	liquidity and crowding; short-horizon reversal	turnover_rate, auction_return
10	d3884de8-ea1	liquidity and crowding; trend or continuation	ret_20d, turnover_rate

7.1 Child-factor dictionary

Child proxy	Economic interpretation
abnormal_vol_rev	abnormal-volume reversal and transient price pressure

amihud_20d	Amihud illiquidity and price impact per traded value
auction_return	auction imbalance and short-lived price pressure
log_dollar_vol	trading-value scale, capacity, and crowding
ret_120d_skip5	persistent trend excluding the most recent trading week
ret_20d	medium-horizon continuation versus reversal state
rev_5d	short-horizon return reversal
turnover_5d	short-horizon liquidity demand
turnover_rate	share-turnover intensity and crowdedness
volume_profile_ratio	relative volume-shape abnormality and intraday participation

8 Threats to validity

1. The adaptive afternoon search inspected 2023 OOS outcomes. A new untouched period is mandatory before making an external performance claim.
2. Reported signal Sharpe excludes commissions, stamp duty, market impact, limit-up/down execution, T+1 constraints, and capacity.
3. The strict portfolio is correlation-diverse in daily PnL, but economic primitives still overlap around liquidity, crowding, continuation, and reversal.
4. Sequential diagnostics within the inspected OOS window show that correlation diversification is not uniformly stable: the peak subperiod maximum pairwise correlation is 0.7409. Formal false-discovery-rate analysis and untouched walk-forward regime validation remain future work.

9 Reproducibility

The final manifest is `paper/afternoon_final_factors.json`. Raw JSONL logs for all three extension stages are stored under `experiment_results/`; exact filenames are listed in the handoff. Descriptive phase and split artifacts are `paper/afternoon_phase_audit.json` and `paper/afternoon_regime_audit.json`. All runs are local research measurements; this workflow performs no external Alpha submission. A static SHA-256 delivery snapshot is written to `paper/afternoon_delivery_index.json`.

10 Untouched-holdout preregistration

The next validation protocol is frozen in `paper/next_holdout_preregistration.json`. It preserves the ten displayed expressions and their order before any new results are opened. The future evaluation must use a consecutive period disjoint from 2020–2023, prohibit holdout-driven replacement or retuning, and report transaction-cost, capacity, block-bootstrap, and multiple-testing diagnostics. The preregistration contains no new performance measurement and must not be represented as validation.

11 Conclusion

The expanded search establishes a measured, auditable set of at least ten factors satisfying the requested Sharpe and correlation constraints. The result is materially stronger than the earlier checkpoint because it adds cross-style ensembles and explicit residualized lineages while preserving negative results. It remains a research checkpoint rather than a top-conference-ready empirical claim until untouched holdout validation and execution modeling are complete.

References

- [1] De Bondt, W. and Thaler, R. (1985). Does the stock market overreact? *Journal of Finance*. [doi](#).
- [2] Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers. *Journal of Finance*. [doi](#).
- [3] Fama, E. and French, K. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*. [doi](#).
- [4] Harvey, C., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*. [doi](#).

- [5] Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*. [doi](#).
- [6] White, H. (2000). A reality check for data snooping. *Econometrica*. [doi](#).
- [7] Hansen, P. (2005). A test for superior predictive ability. *Journal of Business and Economic Statistics*. [doi](#).
- [8] Bailey, D. and López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*. [SSRN](#).
- [9] Frazzini, A., Israel, R., and Moskowitz, T. (2018). Trading costs of asset pricing anomalies. Working paper. [author PDF](#).